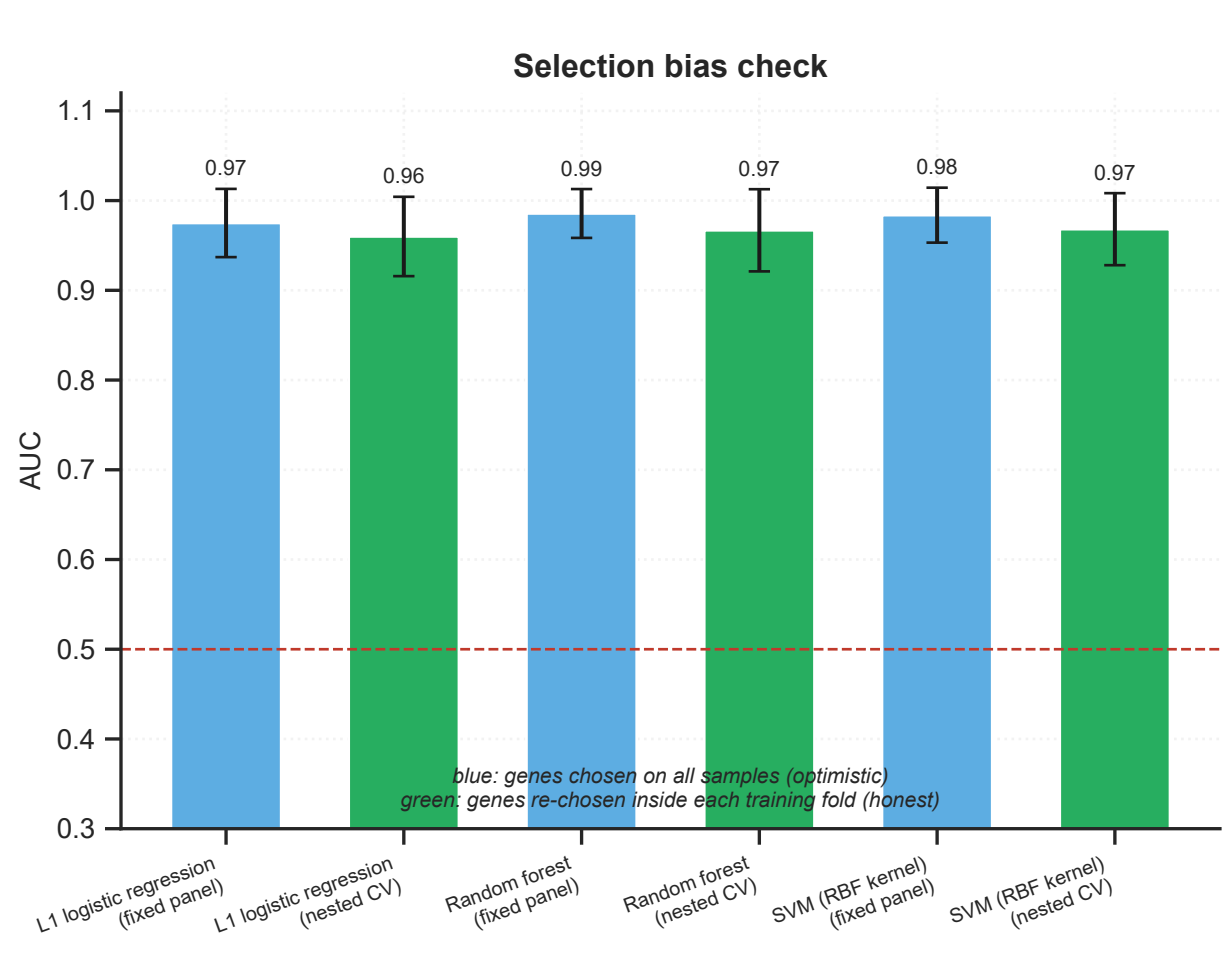
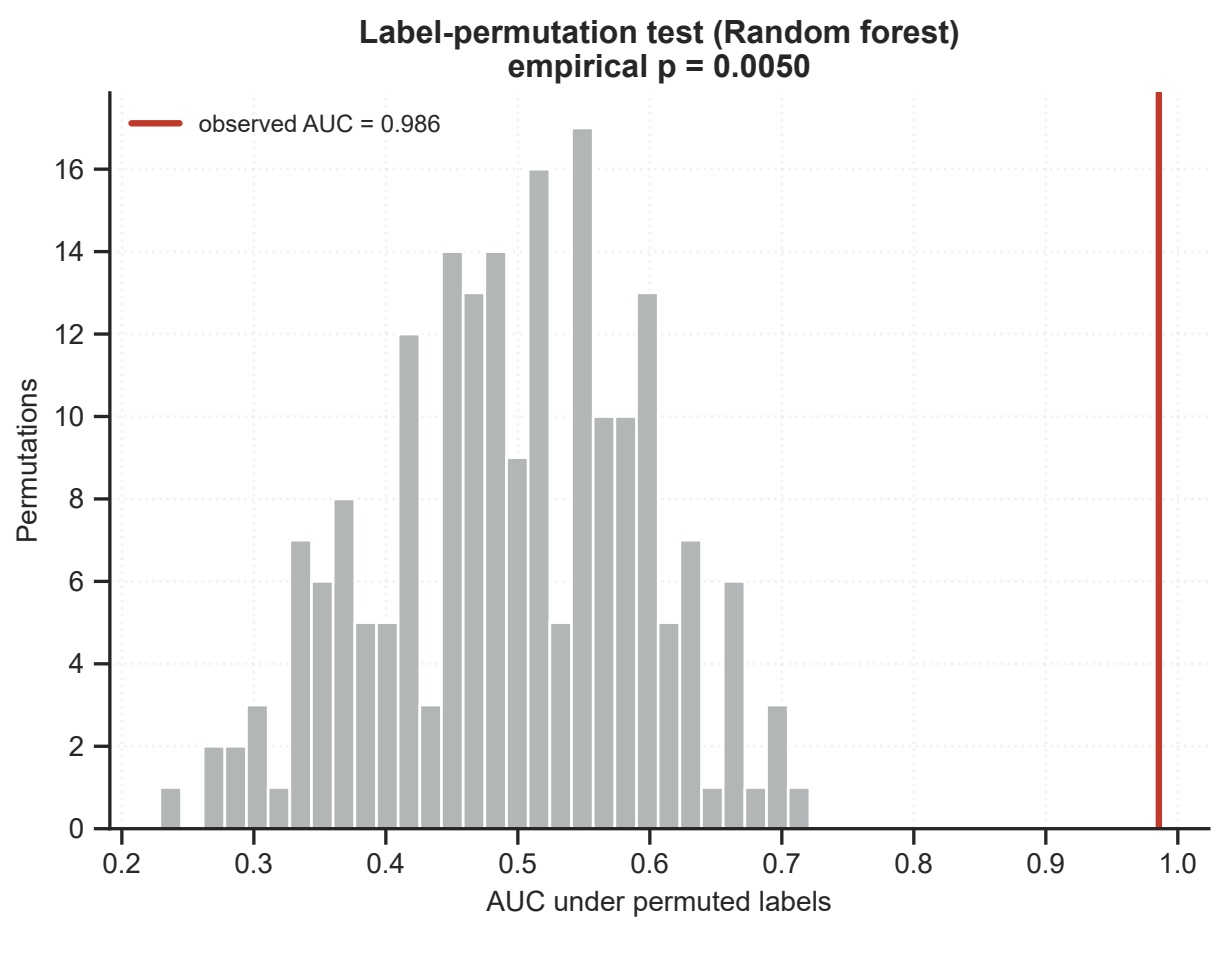

Selection bias check
AUC
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.97
0.96
0.99
0.97
0.98
0.97
blue: genes chosen on all samples (optimistic)
green: genes re-chosen inside each training fold (honest)
L1 logistic regression (fixed panel)
L1 logistic regression (nested CV)
Random forest (fixed panel)
Random forest (nested CV)
SVM (RBF kernel) (fixed panel)
SVM (RBF kernel) (nested CV)


Label-permutation test (Random forest)
empirical p = 0.0050
observed AUC = 0.986
Permutations
16
14
12
10
8
6
4
2
0
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
AUC under permuted labels